

Structural Primers Trigger Procedural Shortcuts in LLM Graph Reasoning

Gal Barak, Nitzan Zacharia, and Inbal Moryles

Tel Aviv University

Abstract

Structural node features help graph learners distinguish relations, but their textual counterparts can also hand a language model the answer to a graph query. We synthesize a graph-blind primer control and a paired degree-neighborhood diagnostic on 84,000 unique Qwen3 responses to 40-node graphs. With degree text, plain Qwen3-4B improves edge counting from 2.00% to 29.25%, yet loses joint degree-neighborhood correctness (85.75% to 76.50%). Its degree responses shift from neighbor enumeration toward retrieval: at density $p = .50$, accuracy falls from 99% to 87%, whereas at $p = .85$ it rises from 33% to 60%. Two-decimal random-walk features leave just 3.30 distinct pairs among 40 nodes at $p = .50$. These results bound the benefit of textual structural features: answer availability can redirect observable solution patterns without ensuring graph-consistent answers, while rounding loses distinctions before inference begins.

1 Introduction

Textual graph evaluations often treat a correct answer as evidence that a model used graph structure. That inference fails when a prompt already states a degree, or when two separately correct-looking answers do not satisfy the graph identity $d_v = |N(v)|$. Graph neural networks (GNNs) use degree, clustering, and random-walk structural encoding (RWSE) as numerical features; serializing these features as sentences changes both what a solver can copy and what distinctions it can observe.

We ask: *How do textual structural primers alter an LLM’s procedural execution of graph tasks, and what are the expressivity limits of these primers in preserving structural invariants?* The gap is between access to graph text and demonstrable algorithmic graph use. A correct `node_degree` response under a degree primer can be a lookup. A correct `edge_count` response can use $|E| = \frac{1}{2} \sum_v d_v$ without consulting adjacency. Accuracy

on either task does not show that the model’s separate neighborhood answer agrees with its degree answer.

We join two diagnostics on the saved prompts and responses: a graph-blind exact rule tests whether the rendered primer determines the target, and paired predictions for a node test agreement (C) and joint correctness (J). Surface response coding distinguishes explicit neighbor enumeration from answers that omit the neighbor list. Finally, we count distinct RWSE strings to bound information available through the feature channel. Across the observed plain 4B cells, degree text raises global edge-count success by 27.25 percentage points (pp) and lowers local J by 9.25 pp. Our testable account is that direct answer access favors a shorter observable response route, which can help when adjacency counting is unreliable and hurt when counting succeeds. This account predicts both a density-dependent sign reversal and no guarantee that separately queried graph identities remain correct.

2 Related Work

GraphQA shows that graph encodings and tasks affect LLM accuracy (Fatemi et al., 2024); NL-Graph and GraphArena extend graph computations (Wang et al., 2023; Tang et al., 2025). These benchmarks chiefly score individual final answers, which cannot separate copying a stated statistic from executing an adjacency procedure or detect inconsistent answers to related queries. A theoretical transformer construction can express degree and edge count (Sanford et al., 2024), but representability alone does not establish that a generated answer used that construction. Work on structural prompts for node classification shows that relevant words can be useful without structural use (Huang et al., 2024). Shortcut learning provides the broader warning that predictive cues need not instantiate the intended computation (Geirhos et al., 2020).

Message-passing GNNs have a 1-WL expressivity bound: the first refinement distinguishes nodes by degree (Xu et al., 2019). Numerical positional and structural encodings, including RWSE, can distinguish additional configurations (Dwivedi et al., 2022; Rampasek et al., 2022). A learned graph-to-language interface trains an encoder for the frozen LLM (Perozzi et al., 2024); our primer instead transfers fixed-precision values as plain sentences. A frozen LLM reading two-decimal strings does not inherit the underlying numerical resolution. Our comparison concerns *node partitions induced by the printed feature*, not the expressivity of a GNN with topology. Because longer context can itself change performance (Shi et al., 2023; Levy et al., 2024), the evaluation includes a content-free filler primer; it is an approximate length control rather than a token-matched placebo.

3 Data and Method

Recorded corpus. The main sweep has independently seeded $G(40, p)$ graphs at $p \in \{.10, .20, .35, .50\}$: 100 graphs per density, six tasks, seven primer conditions, and four Qwen3 arms (1.7B and 4B, plain and native thinking), giving $4 \cdot 4 \cdot 100 \cdot 6 \cdot 7 = 67,200$ outputs. An extension adds $p \in \{.65, .75, .85\}$ for `node_degree` and `edge_existence`, another 16,800. We audited the repository’s Qwen3 `densfull140` and `densfull140hi` run shards at commit `d9b2190`: 84,002 raw lines contain two identical duplicate keys; the 84,000 unique records pair by arm, graph/query instance, task, and condition. The prompts are in the correspondingly named `prompts` files; gold answers match independent reconstruction from the recorded seeds.¹

The six tasks ask for node and edge counts, degree and neighbors of a queried node, edge existence, and a cycle. Degree, neighbors, and edge existence use the same graph/query seed. `node_count` and `cycle_check` have constant gold values on these 700 graphs and are excluded from comparative inference. All prompts use an incident adjacency encoding. Seven conditions are `none`, `filler`, graph-level component count, per-node clustering coefficient, RWSE, degree, and `all` (degree, clustering, RWSE). The filler re-

peats the node set without adding edge facts; clustering and RWSE are printed to two decimals. RWSE consists of (P_{vv}^2, P_{vv}^3) , $P = D^{-1}A$, with zero rows for isolates. The shared renderer is `graphtalk/primers.py`.

For each density p and graph index $i \in \{0, \dots, 99\}$, the builder sets $s = 20260906 + 10^6(1 + \text{round}(1000p)) + 40000 + i$ and draws the undirected graph with that seed; an independently initialized random generator selects the task’s query. The identical prompt records feed all four arms. We compare saved outputs on the same graph/query identifiers under each condition. This is a reanalysis of recorded generations, with no new model outputs or substituted model checkpoints.

Scoring and tests. Greedy generation uses an 8,192-new-token cap in the main sweep; in the high-density extension plain arms have 2,048 and thinking arms 8,192. We score exact integers, exact neighbor sets, and exact booleans; an unparsed or capped answer fails. Each main-sweep cell has $n = 400$ matched outputs. Paired changes are in pp relative to `none`. For the focal J contrast, we use the supplied density-stratified paired bootstrap interval and exact McNemar test, with Benjamini–Hochberg correction within the six primer contrasts for each arm/diagnostic (McNemar, 1947; Benjamini and Hochberg, 1995). Response-style shares use terminated generations only; accuracies include capped failures. An exploratory correction within each family is not a simultaneous guarantee across every result in the paper. A targeted, non-exhaustive audit of 52 extracted neighborhood responses in the source analyses shifted C or J by no more than 1.25 pp in any audited cell; the focal plain-4B cells did not change.

Answer availability on these prompts. An edge-list-blind rule parses only the rendered primer, never the adjacency sentences. For degree, it selects the sentence for the queried node; for edge count, it sums the 40 printed degrees and halves the result. We applied these rules to the saved prompts under `degree` and `all`: each recovers all 700 degree targets (four main and three extension densities) and all 400 edge-count targets (main sweep), separately for each primer condition. These checks use the actual graph/query instances behind the 84,000 outputs. They establish direct answer leakage by an exact identity, independently of whether a

¹Data, renderer, extractor, and audit inputs: <https://github.com/GalBrk/GraphTalk/tree/d9b2190f3ad0e40c0392159839a9c6167132704e>. Only the four Qwen3 arms and the two 40-node corpus families enter this analysis.

model uses it. The clustering and RWSE sentences do not state these targets, but may provide approximate density or degree information; we make no zero-leakage claim for them.

Paired invariant. For the separately prompted degree $\hat{d}_v$ and neighborhood $\hat{N}_v$ on one graph and node, $C_v = 1$ if both responses terminate, parse, contain valid node IDs, and $\hat{d}_v = |\hat{N}_v|$; otherwise $C_v = 0$. Set $J_v = S_{v,\text{degree}} S_{v,\text{neighbors}}$, where S is exact uncapped success. Thus $J \leq C$ and $C - J$ counts consistent but jointly incorrect pairs. This tests an output-level necessary invariant; it does not reveal a latent graph representation.

Observable procedure. For a terminated degree response, *enumeration* requires at least five neighbor IDs in separate numbered or bulleted lines. A *retrieval-style* answer gives a degree without either such a list, a comma-separated run of five IDs, or an explicit claim to have counted connections. Other responses remain unclassified. The accompanying source bundle gives the coding expressions. Retrieval-style is a surface category and does not prove internal copying. The five-entry rule undercounts enumeration at low density; the main response-style comparisons therefore use $p \geq .35$.

4 Results

4.1 An answer-carrying primer changes the route

Table 1 reports absolute exact success rather than completion-conditioned scores. Plain 4B edge counting improves from 2.00% to 29.25% with degree (+27.25 pp, 95% paired CI [22.75, 31.75]; 114 instances fixed, five broken; exact McNemar $p < 10^{-25}$). Yet its degree success falls 99.25% to 92.75% (−6.50 pp). The same degree primer gives 1.7B only +3.00 pp on edge count (1.25% to 4.25%; adjusted $q = .101$), while 1.7B thinking goes from 15.75% to 12.50%. Answer sufficiency therefore does not ensure reliable use of a 40-term sum.

Plain 4B’s degree responses show where this divergence arises (Figure 1). Without a primer, it explicitly enumerates neighbors in 95% and 100% of terminated responses at $p = .35$ and $.50$, respectively, with 99% exact accuracy. With degree, enumeration falls to 6% and 8%, and retrieval-style answers rise to 74% and 62%. At $.50$ accuracy falls 99% to 87% (−12 pp; one corrected, 13 broken; exact McNemar $p = .0018$). At $.85$ it rises 33% to

Arm	Task	None	Deg.	All	$\Delta\text{Deg.}$
1.7B	degree	60.50	68.00	70.75	+7.50
	neighbors	72.50	76.00	74.50	+3.50
	edge count	1.25	4.25	2.00	+3.00
1.7B-T	degree	76.25	85.00	88.50	+8.75
	neighbors	78.75	85.00	85.00	+6.25
	edge count	15.75	12.50	20.75	−3.25
4B	degree	99.25	92.75	91.25	−6.50
	neighbors	86.00	83.00	83.25	−3.00
	edge count	2.00	29.25	5.75	+27.25
4B-T	degree	97.00	99.50	97.50	+2.50
	neighbors	95.50	96.00	94.25	+0.50
	edge count	20.00	28.50	21.75	+8.50

Table 1: Main sweep exact success (% , $n = 400$ per cell, $p \leq .50$). “T” denotes native thinking; capped outputs count as failures. All includes degree, clustering, and RWSE. $\Delta\text{Deg.}$ is Degree minus None, in percentage points.

60% (+27 pp) while 69% of degree-primer outputs are retrieval-style. The sign reverses between the 99% and 33% no-primer regimes. The $.85$ cell also has a shorter plain-arm generation cap and a longer adjacency list, so the two density regimes do not isolate density alone.

Adding more answer-carrying information does not monotonically help: plain 4B’s all primer includes degree but attains only 5.75% edge-count success, compared with 29.25% under degree alone and 2.00% under none. At the same time, degree information is directly retrievable for the local degree query. The −6.50 pp degree effect in Table 1, together with 99.25% no-primer success, rules out an explanation in which supplying a mathematically correct value must weakly improve performance. It also motivates measuring the response route at each density instead of pooling across regimes where baseline counting differs by 66 pp (99% at $.50$ versus 33% at $.85$).

The response-style shift is model-dependent. Under degree, 1.7B thinking enumerates in 76–98% of terminated responses at $p \geq .35$ and gains 8.75 pp on main-sweep degree (76.25% to 85.00%); its seven-density gain is 16.0 pp on exact success with capped outputs counted as errors. Conversely, the 4B thinking arm raises degree from 97.00% to 99.50%. These patterns are consistent with substitution in plain 4B and using the primer as a check in 1.7B thinking; they cannot establish hidden algorithms.

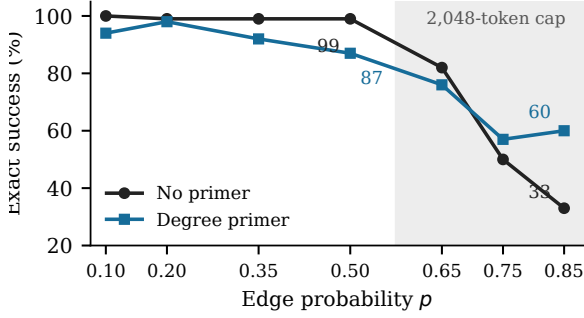


Figure 1: Plain 4B degree exact success across all seven densities (100 graphs each). Shading marks $p \geq .65$, where the plain-arm cap changes from 8,192 to 2,048 new tokens. Response-style percentages appear in the text and are conditional on termination; plotted success counts caps as failures.

4.2 Side information and task prevalence

Exact target leakage constrains what a different primer effect means. On 4B thinking, component and clustering primers each raise edge-count success from 20.00% to 44.75% (+24.75 pp), although neither states the 40 degrees or an exact edge count. No direct edge-count sentence explains this gain. For no primer, the 4B thinking arm terminates 20.50% of edge-count generations and is correct on 97.56% of completions, producing 20.00% success under the cap. With degree text, completion reaches 41.25% while exact success reaches 28.50%. A primer can change completion and answer quality separately; the 44.75% component/clustering effects should not be described as copying an exact edge count.

For `edge_existence`, raw accuracy can increase as true edges become prevalent without better discrimination. In plain 1.7B under none, $p = .50$ has 54% accuracy: 48 true positives and six true negatives, with 46 false positives. At $p = .85$, it has 86% accuracy: 85 true positives, one true negative, and 14 false positives. The model answers “yes” in 94% and 99% of queries, respectively; balanced accuracy is 55.77% and 53.33%. At .85, an always-yes predictor already obtains 85% because 85 of the 100 queried pairs are edges. This result limits claims of improved edge reasoning from a higher raw score at greater density. Across the seven densities the all primer reduces this arm’s false-alarm rate from 51% to 33% (−18.6 pp, paired CI [−24.0, −13.1]), whereas filler raises it to 69% (+18.3 pp, [13.9, 22.7]). These shifts show why an approximate length control and prevalence-

Arm	None		Degree	
	C	J	C	J
1.7B	62.25	54.00	73.00	62.50
1.7B-T	81.00	71.00	81.00	76.50
4B	87.00	85.75	81.00	76.50
4B-T	94.00	93.75	96.00	95.75

Table 2: Paired degree–neighborhood cardinality agreement C and joint exact correctness J (% , 400 nodes per arm and condition, $p \leq .50$).

aware metrics matter even when the main question concerns degree and neighborhoods.

4.3 Global counting improves while a local axiom fails

Table 2 shows the paired negative result. For plain 4B, degree lowers J from 85.75% to 76.50% (−9.25 pp; paired 95% CI [−13.75, −4.75]; 28 nodes fixed, 65 broken; exact McNemar $p = .00016$) and C from 87.00% to 81.00%. Hence the same intervention that adds 27.25 pp to edge counting loses 9.25 pp on related local answers. Filler lowers plain 4B’s J by 13.25 pp, so the degree effect cannot be wholly attributed to content independent of length. For plain 1.7B, degree instead increases J from 54.00% to 62.50%, but $C - J$ grows from 8.25 to 10.50 pp: extra agreement does not imply extra correctness. The 4B thinking arm is near ceiling at baseline ($J = 93.75\%$) and gains 2.00 pp.

4.4 Textual precision limits structural resolution

The prompts print RWSE pairs to two decimals. Counting unique pairs in the 100 prompts per density gives 30.14 per 40-node graph at $p = .10$, 6.74 at .35, 3.30 at .50, and 2.00 at .85. At .50, printed degree has 12.50 distinct values on average, while two-decimal RWSE has 3.30. Reformatting the same underlying features to six decimals yields 40.00 distinct pairs at .50 (39.99 at .85). Identical printed pairs make nodes indistinguishable *through this feature field*; adjacency sentences and IDs remain available. This is a measured encoding resolution bound, not a measured accuracy gain under a six-decimal LLM prompt or proof of a global GNN expressivity ordering.

More formally, let $f_h(v)$ be the RWSE pair rounded to h decimals, and partition V by equality of f_h . Any solver restricted to this field must assign the same representation to nodes in one equivalence

class. At $p = .50$, the two-decimal partition has on average 3.30 cells; degree alone has 12.50, while the six-decimal RWSE counterfactual has 40.00. The most common two-decimal pair is shared by 62.68% of nodes on average at that density. The equality constraint is imposed by the serialization step, before any model response or decoding cap. It does not assert that the full prompt cannot resolve the nodes: their adjacency lines and numerical IDs supply other channels.

5 Discussion and Conclusion

The two diagnostics separate answer access from its consequences. Degree text is an exact side channel for local degree and global edge count; in plain 4B it co-occurs with a 27.25-pp global gain, a 6.50-pp degree loss, and a 9.25-pp drop in local joint correctness. Its visible replacement of enumeration is harmful when no-primer counting reaches 99% at $p = .50$ and helpful when that baseline reaches only 33% at $p = .85$. Thus a positive task-level score is insufficient evidence of general graph understanding. Evaluations should disclose primer-only recoverability, density-conditioned baselines, paired invariants C and J , and the number of distinguishable printed feature values.

The diagnostic pair also distinguishes two kinds of negative result. In plain 4B under none, $C - J = 1.25$ pp (87.00% minus 85.75%); under degree, it grows to 4.50 pp (81.00% minus 76.50%). The degree primer creates both more outright violations of $\hat{d}_v = |\hat{N}_v|$ (19.00% versus 13.00%) and more self-consistent yet jointly wrong outputs. This is stronger evidence against a uniform local benefit than the -6.50 -pp degree-task change alone, because J requires two independently queried answers to match the same graph. Yet even $C = 1$ would certify only a cardinality relation: a degree of 20 and an invented 20-element neighbor set satisfy the equation. Both C and J are therefore needed.

Across arms, the direction of the effect depends on which route remains reliable. Plain 1.7B begins at 60.50% degree and 72.50% neighbor success and gains 7.50 and 3.50 pp with degree text, while its J gains 8.50 pp; its low no-primer degree baseline leaves room for help. Plain 4B begins at 99.25% degree and 86.00% neighbor success and loses 6.50 and 3.00 pp, with a 9.25-pp loss in J . Thinking 4B begins at $J = 93.75\%$ and changes to 95.75%. Pooling the four arms would obscure

that the largest positive edge-count response and the clearest negative invariant response occur in the *same* plain-4B arm. The result supports a conditional procedure-substitution account rather than a single “primers help” coefficient.

The reported contrasts are descriptive of this fixed interface. Filler costs plain 4B 12.50 pp on neighbors (86.00% to 73.50%) and 13.25 pp on J , which precludes assigning all of degree’s -9.25 pp in J to semantic content alone. Conversely, component count improves plain 4B neighbors by 9.25 pp (86.00% to 95.25%) without stating any local membership list. These non-answer effects motivate token-matched controls in a future intervention; the current data do not estimate a content-only causal effect. The narrower observation survives that distinction: the actually deployed, answer-carrying degree text raises global edge-count success and lowers joint local correctness under the recorded prompts.

Our scope is four arms of one model family, one graph family, one text ordering, and saved deterministic outputs. The high-density plain-arm cap differs from the main sweep; filler is not token matched. Style codes describe response text, and extracted neighbor sets have a limited manual parser audit rather than exhaustive validation. Feature collisions describe the RWSE field only. These constraints preserve the central conclusion: structural primers can induce brittle observable shortcuts without guaranteeing graph-consistent procedural execution.

The next experiment should cross degree content with a token-matched, numerically shuffled control under identical generation caps at all densities, then test whether the -9.25 -pp plain-4B J change persists beyond filler and cap effects. A second experiment should give the models two-, four-, and six-decimal RWSE at matched length and score node-pair discrimination alongside C/J : the observed 3.30 versus 40.00 feature classes at $p = .50$ are a prediction about available information, not yet an LLM accuracy result. Finally, re-label nodes and evaluate balanced edge queries on degree-regular and block-model graphs, measuring exact answer leakage and C/J again. These tests would determine whether the observed route shift extends beyond the present $G(40, p)$ interface.

References

- Yoav Benjamini and Yoel Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. Talk like a graph: Encoding graphs for large language models. In *International Conference on Learning Representations*.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Jin Huang, Xingjian Zhang, Qiaozhu Mei, and Jiaqi Ma. 2024. Can llms effectively leverage graph structural information through prompts, and why? *Transactions on Machine Learning Research*.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15339–15353.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.
- Ladislav Rampasek, Mikhail Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. 2022. Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*.
- Clayton Sanford, Bahare Fatemi, Ethan Hall, Anton Tsitsulin, Mehran Kazemi, Jonathan Halcrow, Bryan Perozzi, and Vahab Mirrokni. 2024. Understanding transformer reasoning capabilities via graph algorithms. In *Advances in Neural Information Processing Systems*.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 31210–31227.
- Jianheng Tang, Qifan Zhang, Yuhao Li, Nuo Chen, and Jia Li. 2025. Grapharena: Evaluating and exploring large language models on graph computation. In *International Conference on Learning Representations*.
- Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. 2019. How powerful are graph neural networks? In *International Conference on Learning Representations*.